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Trends and Disparities in Unmet Treatment Needs for Co-Occurring Depression and Alcohol Use Disorders Among Young Adults in the U.S.

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Major depressive episode (MDE) and alcohol use disorder (AUD) often co-occur and the presence of one doubles the risk for the other, which brings disabling social and health consequences. Increasing evidence supports integrated treatment of co-occurring MDE and AUD, including combined medications for both conditions as well as behavioral therapies. While young adults suffer disproportionately from these co-occurring conditions, little is known about their treatment utilization. Using nationally representative data from the National Survey on Drug Use and Health (NSDUH), 2011–2019, this study examined temporal trends and disparities in the prevalence and treatment use for co-occurring MDE and AUD among young adults aged 18–25 in the U.S. Across the survey time, an increasing trend of MDE was revealed, whereas the prevalence of AUD decreased significantly. Meanwhile, the prevalence of co-occurring MDE and AUD remained steady. Among young adults with co-occurring MDE and AUD, while treatment use for MDE increased from 2011 to 2019, treatment use for AUD and co-occurring conditions remained stable. Observed widening disparities in co-occurring treatments utilized among young adults ages 18–25 are further perpetuated by gender, emerging adulthood, and socioeconomic status, warranting immediate action.

Public Policy Relevance Statement

Young adults with co-occurring depression and alcohol disorders suffer from serious physical health, economic, and social consequences that can last a lifetime if left untreated. Examining trends regarding the prevalence of these conditions among young people 18–25 and their treatment use patterns reveals the unique needs of this population. These findings may enhance practice and policy innovations that conceivably could help to mitigate against widening disparities.

Major depressive episode (MDE) and alcohol use disorder (AUD) are two of the top five leading causes of death in developed countries, with young people bearing a disproportionate burden (Deady et al., 2016). MDE has been a key risk factor for suicidal behavior in young adults (Furr et al., 2001). Similarly, young adults with AUD are three to four times

more likely to attempt suicide compared to those without AUD (Esposito-Smythers & Spirito, 2004). MDE and AUD often co-occur and encompass one of the most common and disabling psychiatric conditions (Conner et al., 2009; Hasin et al., 2007; Schuckit, 2006; Turner & McLellan, 2009). In a meta-analysis of epidemiological studies, the presence of one disorder doubled the risk of having the other, irrespective of measurement differences (Boden & Fergusson, 2011). In terms of pathways, AUD may precede MDE, MDE may precede AUD, or their reciprocal relationship may increase risk for the other. Boden and Fergusson (2011) theorize AUD increases risk of MDE via metabolic changes (McEachin et al., 2008), socioeconomic mechanisms (Caetano et al., 2001), and genetic factors (Fu et al., 2002; Kuo et al., 2010; Luo et al., 2005). Conversely, the self-medication hypothesis posits the

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opposite directionality, whereby alcohol is misused to alleviate symptoms of depression (Bolton et al., 2009). Indeed, elevated depressive symptoms are related to more isolated and excessive drinking (Keough & O'Connor, 2014) as symptoms like poor motivation and lack of interest increase time spent alone, which consequently enhances the need to ease the loneliness (Baddeley et al., 2013). Regardless of the potential causal direction of these co-occurring conditions, their harmful health, economic and social consequences are evident in the greater severity and persistence, suicide risk, and lower social adjustment associated with their co-occurrence (Boschloo et al., 2011; Cornelius et al., 1995; Grant et al., 1996; Salloum et al., 1995). It is known that most mental health disorders have onset during young adulthood by age 25 (Pedrelli et al., 2015) and that the two most common conditions are alcohol misuse and depressive symptoms (Pedrelli et al., 2016). Although epidemiological studies have reported the characteristics and rates of co-occurring MDE and AUD (Boschloo et al., 2011; Grant & Harford, 1995; Hasin et al., 2007), little is known about the scope and patterns of these co-occurring conditions in young adults. Such knowledge is critical to inform targeted intervention efforts for young adults with both conditions.

While extant literature reveals patterns of service use for these conditions separately, nascent research exists about treatment use among young adults with both conditions. In a national sample, the proportion of young adults with MDE who received MDE treatment increased from 28.2% to 29.7% between 2005 and 2014 (Mojtabai et al., 2016). We also know from a national sample that young adults with AUD who received AUD treatment were 7.7% in 2002 (Wu et al., 2007). Although evidence supports the benefits of simultaneous treatment of co-occurring MDE and AUD (Kay-Lambkin et al., 2009; Pettinati & Dundon, 2011), there are continuing debates regarding antidepressant medication interactions with alcohol misuse, including claims that depressive symptoms induced by substance use do not represent a true underlying MDE (Pettinati & Dundon, 2011). However, the efficacy of integrated treatments and the case that both disorders be treated concurrently has become increasingly warranted as findings demonstrate those abstaining from alcohol are more vulnerable for eventual relapse due to persisting depression (Pettinati & Dundon, 2011). There is evidence that combined use of medications for alcohol misuse (i.e., naltrexone) and for depression (i.e., sertraline) may be the optimal course of treatment (Pettinati et al., 2010). Additionally, emerging findings indicate psychological interventions, such as motivational interviewing (MI) and cognitive behavioral therapies (CBT), can be effective in significantly reducing both alcohol consumption and depressive symptoms (Baker et al., 2010, 2012; Kay-Lambkin et al., 2009). Yet, more research is needed about the prevalence of treatment utilization and related sociodemographic disparities for young adults with these co-occurring conditions.

Based on the conceptual framework of Social Determinants of Health, individuals' health outcomes are influenced by their social and economic status (Kneipp et al., 2018; World Health Organization [WHO], 2008). Previous research further suggests that the prevalence and treatment use for MDE and AUD among young adults differ by demographic (e.g., age, gender, race) and socioeconomic factors (e.g., insurance status, income; Weitzman, 2004). To expand findings from previous studies, our study also included these variables to examine potential sociodemographic disparities in young adults' unmet treatment needs for co-occurring

MDE and AUD. Understanding these patterns of service use among young people is critical to inform treatment engagement initiatives and can support providers and policy makers in tailoring diverse public health campaigns that are particularly impactful for this heterogeneous population.

The purpose of this epidemiological study was to obtain a large-scale perspective on one of the biggest public mental health crises among young adults by tracking changes in MDE and AUD comorbidity and treatment utilization over the span of 9 years. Specifically, using nationally representative data for young adults ages 18–25 from the National Survey on Drug Use and Health (NSDUH), 2011–2019, we examined the temporal trends in the prevalence and treatment of co-occurring MDE and AUD and identified disparities in unmet treatment needs among young adults with both conditions. While this study was essentially descriptive in nature, we anticipated to observe:

1. a significant increase in MDE prevalence but decrease in AUD prevalence over time (Hypothesis 1);
2. an increasing number of young adults with co-occurring MDE and AUD who have received treatment for MDE, AUD, and both conditions across the survey years (Hypothesis 2); and
3. significant differences in the prevalence and treatment of co-occurring MDE and AUD by young adults' sociodemographic characteristics (Hypothesis 3).

Method

Sample

The NSDUH is an annual cross-sectional survey sponsored by the Substance Abuse and Mental Health Services Administration within the U.S. Department of Health and Human Services. Using a stratified multistage area probability sampling method, the NSDUH provides nationally representative data for the civilian, noninstitutionalized U.S. population aged 12 or older from all 50 states and the District of Columbia (Substance Abuse and Mental Health Services Administration [SAMHSA], 2020). For this study, publicly available data for young adults aged 18–25 for 2011–2019 surveys were analyzed. The survey is administered in English and Spanish using computer-assisted interviewing.

Measures

Major Depression. Major depression in the NSDUH was measured using a structured interview based on the fourth edition of the Diagnostic and Statistical Manual of Mental Disorders (DSM-IV; American Psychiatric Association [APA], 1994). An adult participant was classified as having a 12-month major depressive episode (MDE) if they reported having experienced at least five out of nine criteria that define MDE in the prior year, where at least one criteria is depressed mood or loss of interest or pleasure in daily activities.

Adult participants with 12-month MDE were further asked questions from the Sheehan Disability Scale to measure the level of MDE-related functional impairment in four major life activities or role domains (i.e., home management, ability to work, close relationships, and social life). Ratings of seven and above were considered severe impairment on a zero to ten visual analog scale with

categories of “no interference” (0), “mild” (1–3), “moderate” (4–6), “severe” (7–9), and “very severe” (10). For this study, we dichotomized participants as having MDE-related severe functioning impairment (7–10) or not (1–6).

Alcohol Use Disorders. In the NSDUH survey, participants were asked about their use of alcohol in the past year. If respondents reported using a substance in the past year and on more than 5 days for alcohol, they were further asked questions that correspond to the DSM-IV criteria for dependence or abuse (e.g., tolerance, withdrawal, taking larger amounts or taking them for longer periods, inability to cut down, time spent using the substance, giving up activities, and continued use despite problems) in the past year. Participants were categorized as having 12-month AUD if they met DSM-IV criteria for alcohol dependence or abuse.

Treatment. To assess treatment for depression, adults with 12-month MDE were asked if they had received any inpatient, outpatient, or prescription medication treatment for their MDE. To assess treatment for AUD, adult participants were asked if they had received treatment or counseling designed to help reduce or stop alcohol use at any location in the past 12 months, including a hospital (in- or outpatient), a rehabilitation facility (in- or outpatient), a mental health center, an emergency room, private doctor’s office, prison or jail, or self-help group. Participants were considered as receiving treatment for MDE and AUD if they responded “yes” to both questions.

Sociodemographic Characteristics. Sociodemographic variables examined in this study included participants’ age group (18–21, 22–25), gender (male, female), race/ethnicity (white, Hispanic, Black, Asian/Native Hawaiian or other Pacific Islanders, and other), insurance coverage (uninsured, insured), and annual household income (less than \$20,000–\$75,000 or more).

Data Analysis

The analysis was conducted in three stages. First, simple logistic regression analyses were conducted to assess time trends in the prevalence of MDE, AUD, and their co-occurrence; Hypothesis 1 was tested, and odds ratios (OR) with 95% confidence intervals (CI) were reported. Second, among young adults with co-occurring MDE and AUD, simple logistic regression models were further used to examine trends and patterns in their treatment use for MDE, AUD, and co-occurring conditions, respectively, and Hypothesis 2 was tested. In all simple regression models, survey year was entered as the continuous independent variable. Lastly, to test Hypothesis 3, a series of multivariable logistic regression analyses were conducted to identify sociodemographic differences in the prevalence and treatment of MDE, AUD, and co-occurring conditions. To adjust for potential time influence, the survey year variable was included in all multivariate models. Participants’ MDE-related severe impairment was further added in the multivariate models for treatment and adjusted odds ratios (AOR) were reported. Missing data ranged from 0% to 3.7% for the variables included in these analyses and were removed with the use of listwise deletion (Dong & Peng, 2013). All analyses were performed using R, Version 4.0.3, adjusting for sampling weights provided by the NSDUH (R Core Team, 2020). The Institutional Review Board at Research Triangle

Institute (RTI) International approved the NSDUH data collection protocol.

Results

In total, 207,895 young adults participated in the NSDUH 2011–2019. As listed in Table 1, participants with both conditions were mostly females (56.3%), whites (65.4%), and aged 22–25 (73.1%). Around 20.5% did not have any insurance, and 26.4% came from households with annual income below \$20,000.

Trends in the Prevalence and Treatment of MDE, AUD, and Co-Occurring Conditions

As displayed in Figure 1, a steady increase was observed in the prevalence of 12-month MDE among all participants, from 8.0% in 2011 to 13.0% in 2019 ($OR = 1.07, P < .001$). In contrast, the prevalence of AUD decreased from 12.2% in 2011 to 8.8% in 2019 ($OR = .95, P < .001$). Overall, the annual prevalence of co-occurring MDE and AUD remained relatively stable at between 2.5% and 3.4%.

Figure 2 shows the annual prevalence of young adults who reported having received treatment for MDE only, AUD only, and both conditions at any location in the past year among those with co-occurring conditions. In 2011, 44.6% of participants with both conditions received treatment for MDE only, and the receipt increased significantly to 52.3% in 2019 ($OR = 1.03, P < .05$). Likewise, the receipt of prescription medication for MDE increased significantly from 34.4% in 2011 to 38.7% in 2019 among participants with co-occurring MDE and AUD. Meanwhile, the degree of treatment use for AUD remained largely stable at between 9.5% and

Table 1

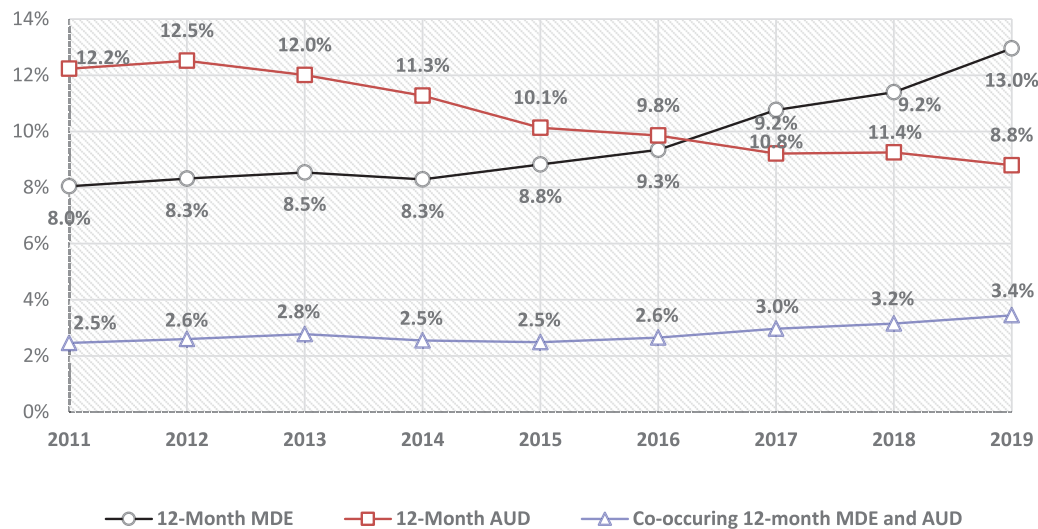
Sociodemographic Characteristics of Young Adults ($N = 207,895$) and Those With Co-Occurring 12-Month MDE and AUD in the NSDUH ($N = 4,393$; 2011–2019)

Sociodemographic characteristics	All participants (%)	Participants with co-occurring MDE and AUD (%)
Gender		
Male	49.8	43.7
Female	50.2	56.3
Age		
18–21	23.8	26.9
22–25	76.2	73.1
Race		
White	56.0	65.4
Hispanic	21.0	18.0
Black	13.5	8.2
Asian/NHPIs	6.9	4.4
Other	2.6	4.0
Insurance coverage		
No	19.1	20.5
Yes	80.9	79.5
Household income, \$		
<20,000	22.5	26.4
20,000–49,999	33.1	34.4
50,000–74,999	16.2	15.5
≥75,000	28.2	23.7

Note. Weighted percentages are reported. MDE = major depressive episode. AUD = alcohol use disorders. NHPIs = Native Hawaiian or other Pacific Islanders.

Figure 1

Trends in the 12-Month Prevalence of MDE, AUD, and Co-Occurring MDE and AUD Among Young Adults in the NSDUH, 2011–2019



11.7% (OR = 0.98, $P = .56$). The prevalence of treatment use for both conditions also remained stable at below 9% across the survey time (OR = .99, $P = .81$).

Disparities in the Prevalence of MDE, AUD, and Co-Occurring Conditions. As shown in Table 2, females were more likely than males to have MDE only (AOR = 1.86, $P < .001$), less likely to have AUD only (AOR = 0.61, $P < .001$), and more likely to have both conditions (AOR = 1.28, $P < .001$). Compared with young adults aged 18–21, lower prevalence of MDE (AOR = 0.75, $P < .001$) and co-occurring MDE and AUD (AOR = .86, $P < 0.05$) were observed in those aged 22–25. Consistently,

lower prevalence of MDE, AUD, and co-occurring conditions were noted among Hispanics, Blacks, and Asian/NHPIs than whites. Relative to uninsured participants, those who had insurance coverage were less likely to have AUD (AOR = 0.89, $P < .001$). Lastly, lower prevalence of MDE and co-occurring MDE and AUD were found in participants with annual household income above \$20,000.

Disparities in the Treatment for MDE, AUD, and Both Conditions. As listed in Table 3, among participants with co-occurring conditions, females were more likely than males to receive treatment for MDE (AOR = 1.83, $P < .001$) but less likely to receive treatment for AUD (AOR = 0.62, $P < .01$), or both

Figure 2

Trends in Treatment Use Among Young Adults With Co-Occurring 12-Month MDE and AUD

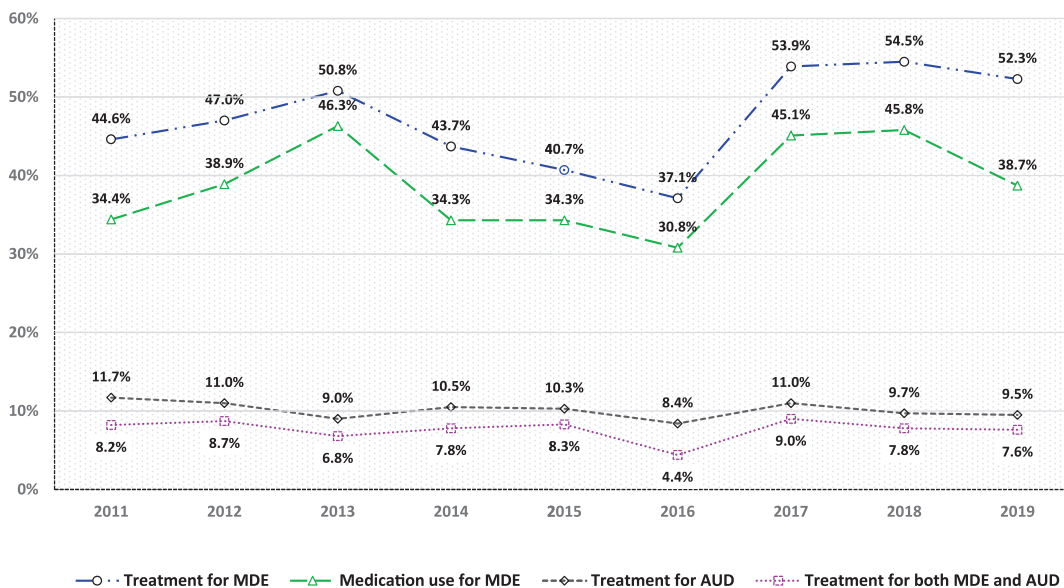


Table 2

Multivariate Differences in the Prevalence of 12-Month MDE (N = 21,012), AUD (N = 22,810), and Co-Occurring MDE and AUD (N = 4,393) Among Young Adults in the NSDUH, 2011–2019

Sociodemographic characteristics	MDE		AUD		MDE and AUD	
	%	AOR (95% CI)	%	AOR (95% CI)	%	AOR (95% CI)
Year	—	1.08 (1.07, 1.09)***	—	0.95 (0.94, 0.96)***	—	1.02 (1.00, 1.04)
Gender						
Male	7.0	(ref.)	12.9	(ref.)	1.7	(ref.)
Female	12.2	1.86 (1.78, 1.94)***	8.2	0.61 (0.58, 0.63)***	2.2	1.28 (1.17, 1.41)***
Age						
18–21	11.5	(ref.)	10.5	(ref.)	2.2	(ref.)
22–25	9.0	0.75 (0.72, 0.79)***	10.6	1.04 (0.99, 1.08)	1.9	0.86 (0.78, 0.93)*
Race						
White	11.2	(ref.)	11.7	(ref.)	2.3	(ref.)
Hispanic	7.4	0.59 (0.55, 0.62)***	10.1	0.82 (0.78, 0.87)***	1.7	0.68 (0.60, 0.78)***
Black	7.0	0.53 (0.49, 0.57)***	8.1	0.65 (0.60, 0.69)***	1.2	0.47 (0.40, 0.55)***
Asian/NHPIs	7.1	0.59 (0.53, 0.65)***	6.6	0.54 (0.49, 0.60)***	1.3	0.53 (0.42, 0.68)**
Other	13.0	1.08 (0.98, 1.19)	12.5	1.08 (0.98, 1.19)	3.0	1.23 (1.02, 1.49)*
Insurance						
No	9.1	(ref.)	12.0	(ref.)	2.1	(ref.)
Yes	9.7	0.95 (0.87, 1.01)	10.2	0.89 (0.84, 0.94)***	1.9	0.87 (0.76, 1.00)
Family income, \$						
<20,000	11.1	(ref.)	11.3	(ref.)	2.3	(ref.)
20,000–49,999	9.9	0.91 (0.86, 0.96)***	10.4	0.81 (0.26, 2.47)	2.1	0.88 (0.78, 0.98)*
50,000–74,999	8.9	0.79 (0.73, 0.84)***	10.3	1.13 (0.21, 5.90)	1.9	0.79 (0.68, 0.92)**
≥75,000	8.6	0.72 (0.67, 0.76)***	10.4	0.46 (0.15, 1.41)	1.7	0.68 (0.60, 0.77)***

Note. MDE = major depressive episode. AUD = alcohol use disorders. AOR = multivariate adjusted odds ratio. Ref. = reference group; NSDUH = National Survey on Drug Use and Health. Condition categories (i.e., MDE, AUD, and co-occurring MDE and AUD) are not mutually exclusive. All variables listed were included in the multivariable model to predict the prevalence of MDE, AUD, and co-occurring MDE and AUD. Weighted percentages are presented. Values in bold indicate statistical significance.

* $p < .05$. ** $p < .01$. *** $p < .001$.

conditions (AOR = 0.61, $P < .01$). Consistently, higher use of treatment was found for MDE (AOR = 1.39, $P < .001$), AUD (AOR = 2.05, $P < .001$), and both conditions (AOR = 2.52, $P < .001$) among participants aged 22–25 than those aged 18–21. Having insurance significantly increased the odds that participants used treatment for MDE (AOR = 1.39, $P < .05$), but not for AUD or co-occurring conditions. Compared with participants with annual household income below \$20,000, those who were from families with household income of \$20,000–\$49,999 and above \$75,000 were less likely to use treatment for AUD and co-occurring conditions. Lastly, having MDE-related severe impairment significantly increased the odds that participants received treatment for MDE (AOR = 2.51, $P < .001$), AUD (AOR = 2.07, $P < .001$), and for both conditions (AOR = 2.49, $P < .001$).

Discussion

Prevalence Trends

Overall, this study found three divergent trends in the prevalence of MDE and AUD between 2011 and 2019. First, in support of our Hypothesis 1, a steady increase in the prevalence of MDE was observed across survey years. This increasing prevalence has been corroborated in other disciplines attributing rising depression among young people to adoption of smartphones (Lin et al., 2016; Twenge & Campbell, 2019) and other “modern-times” lifestyle changes impacting obesity, malnutrition, sleep and sunlight deprivation, and isolation (Hidaka, 2012). Further consistent with our hypothesis, AUD decreased significantly, which may be explained by increased use

of other substances as potential substitutes for alcohol misuse. Particularly, legalization of marijuana use parallels increasing marijuana use disorders (Hasin et al., 2015; Ryan-Ibarra et al., 2015) and as many as 30% of incoming college freshmen report lifetime use of marijuana (Suerken et al., 2014). Likewise, the use of prescription and illicit opioids have surged as an epidemic due to liberalization of laws by state medical boards for treatment of chronic pain (Manchikanti et al., 2012). This may potentially impact decreased prevalence of AUD among young people, wherein previous alcohol use has been associated with subsequent abuse of prescription opioids in young men (Fiellin et al., 2013), suggesting the possibility for opioid abuse to replace some proportion of previous alcohol abuse. To confirm this assumption, future studies are needed to examine trends of MDE and co-occurring use of substances other than alcohol. Lastly, prevalence of co-occurring MDE and AUD remained largely stable at below 3.4% between 2011 and 2019. It is unclear based on the scope of this study whether this trend was a continuation from prior to 2011 or a new-found plateau.

Treatment Trends

Our findings also revealed trends in treatments of MDE and AUD among those with co-occurring conditions between 2011 and 2019. First, in partial support of our Hypothesis 2, use of MDE treatment—including medication use—increased significantly among those with both conditions. This is promising and suggests the effectiveness of various screening interventions in improving early detection for MDE among young adults in recent years, prompted by depression screening recommendations of the U.S. Preventive Services Task

Table 3*Multivariate Differences in Treatment Use of Young Adults With Co-Occurring MDE and AUD in the NSDUH, 2011–2019 (N = 4,393)*

Sociodemographic characteristics	Treatment for MDE ^a		Treatment for AUD ^b		Treatment for both MDE and AUD	
	%	AOR (95% CI)	%	AOR (95% CI)	%	AOR (95% CI)
Year	—	1.03 (1.00, 1.07)	—	0.98 (0.92, 1.05)	—	0.99 (0.92, 1.07)
Gender						
Male	39.7	(ref.)	12.6	(ref.)	9.6	(ref.)
Female	53.6	1.83 (1.50, 2.22)***	8.2	0.62 (0.33, 2.01)**	6.1	0.61 (0.43, 0.88)**
Age						
18–21	40.0	(ref.)	6.3	(ref.)	4.1	(ref.)
22–25	48.9	1.39 (1.16, 1.68)***	11.5	2.05 (1.45, 2.90)***	8.9	2.52 (1.69, 3.75)***
Race						
White	53.2	(ref.)	11.0	(ref.)	9.0	(ref.)
Hispanic	35.1	0.47 (0.36, 0.61)***	9.0	0.83 (0.54, 1.27)	5.1	0.57 (0.34, 0.96)*
Black	38.3	0.51 (0.38, 0.68)***	6.7	0.56 (0.32, 0.99)*	3.6	0.36 (0.17, 0.77)**
Asian/NHPIs	34.5	0.43 (0.25, 0.73)***	7.2	0.67 (0.21, 2.15)	5.3	0.62 (0.14, 2.70)
Other	44.9	0.65 (0.45, 0.94)	10.6	0.94 (0.55, 1.60)	6.6	0.69 (0.38, 1.27)
Insurance						
No	39.9	(ref.)	11.7	(ref.)	8.3	(ref.)
Yes	49.5	1.39 (1.05, 1.84)*	9.7	0.94 (0.64, 1.34)	7.4	1.03 (0.67, 1.60)
Family income, \$						
<20,000	50.7	(ref.)	13.1	(ref.)	10.2	(ref.)
20,000–49,999	44.7	0.78 (0.61, 0.99)*	8.3	0.55 (0.38, 0.81)**	5.7	0.48 (0.31, 0.73)***
50,000–74,999	46.0	0.85 (0.63, 1.15)	10.6	0.70 (0.42, 1.18)	8.2	0.68 (0.37, 1.25)
≥75,000	49.2	0.88 (0.68, 1.14)	9.1	0.55 (0.38, 0.81)**	7.1	0.58 (0.36, 0.95)*
MDE-related severe impairment						
No	32.7	(ref.)	6.1	(ref.)	3.9	(ref.)
Yes	54.6	2.51 (2.04, 3.10)***	12.0	2.07 (1.43, 2.99)***	9.3	2.49 (1.55, 3.98)***

Note. MDE = major depressive episode. AUD = alcohol use disorders. Weighted percentages are presented. AOR = multivariate adjusted odds ratio. Ref. = reference group; NSDUH = National Survey on Drug Use and Health. All variables listed were included in the multivariable model to predict treatment use for MDE only, AUD only, and both conditions. Treatment categories are not mutually exclusive. Values in bold indicate statistical significance.

^a Participants were asked if they had seen or talked to a medical doctor or other professionals about their MDE symptoms in the past 12 months. ^b Participants were asked if they had used treatment or counseling for alcohol use at any location designed in the past 12 months.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Force (USPSTF) where an affirmative response to just two questions suggested need for further evaluation (Sharp & Lipsky, 2002). Moreover, as use of antidepressants has become the most prescribed class of medications in the U.S., its prescription has become broadly accepted in the general medical sector (Olfson & Marcus, 2009). Indeed, public perception of antidepressant use has normalized largely due to marked increase in spending devoted to direct-to-consumer advertising (Donohue et al., 2007; Shapiro, 2020). In general, antidepressants are more effective than placebo in reducing symptoms of depression among those with co-occurring MDE and AUD (McHugh & Weiss, 2019), but more research is needed to study the comparative effectiveness of active treatments. In addition, some evidence suggests antidepressant medication is effective in mitigating quantity of alcohol use, although *sustained* abstinence or remission are often low (Nunes & Levin, 2004). In a review of 33 studies of antidepressant use among those with co-occurring MDE and AUD, evidence for efficacy of antidepressants on alcohol consumption was deemed low quality, due to high bias in some studies as well as heterogeneity found in both medication type and outcome measures employed, yielding mixed results that warrant further research (Agabio et al., 2018).

Nevertheless, increase in AUD treatment was significantly less among those with co-occurring conditions in comparison to MDE treatment use. Another possible explanation for disproportionate use of MDE treatments compared to AUD treatments among young

adults may be due to uptick in depression screening programs implemented across university campuses in response to this epidemic (Garlow et al., 2008; Hill et al., 2015), prompting more young adults to seek MDE treatment than AUD treatment. Other possible explanations can be found in a national survey of over 11,000 young adults 18–22 years of age citing barriers to AUD treatment use which included unreadiness to stop drinking, unable to afford treatment, fear of being perceived negatively for utilizing treatment, and not knowing where to access treatment (Wu et al., 2007). It is worth mentioning that the commonly used 12-step model of many AUD programs can be at odds with young adults in terms of average age of users (48 years), difficulty relating to primary concerns of predominant participants (i.e., marriage, children, and employment issues), and lower addiction severity than older program participants (less frequency, dependence, medical complications, and withdrawal; Kelly et al., 2008).

To improve treatment use for AUD among young adults, harm reduction programs and peer recovery programs should be more widely implemented, as these are more aligned with the unique developmental tasks associated with emerging adulthood where abstinence is typically not yet warranted and education-only programs do not prohibit them from testing out drinking behaviors (Logan & Marlatt, 2010). Indeed, school-based programs for adolescents like Integrated School- and Community-Based Demonstration Intervention Addressing Drug Use among Adolescents

(Poulin & Nicholson, 2005) and School Health and Alcohol Harm Reduction Project (McBride et al., 2004) have been shown to be more effective in prevention of alcohol misuse than abstinence programs like Project DARE (Drug Abuse Resistance Education), which produced either no effects or potentially harmful effects among youth (Lilienfeld, 2007; Lynam et al., 1999). Among college age adults, Alcohol Skills Training Program (ASTP) and Brief Alcohol Screening and Intervention for College Students (BASICS) have had favorable outcomes, including reduced alcohol intake, in at least two studies (National Institute on Alcohol Abuse and Alcoholism [NIAAA], 2002). Efforts should be targeted toward implementing harm reduction-based programs broadly in educational institutions to empower young people with important information and support.

Notably, across survey years, less than 9% of young adults with co-occurring MDE and AUD used treatments for both conditions and, contrary to our hypothesis, no significant change was observed in the proportion of young adults who used simultaneous treatment for both conditions across the survey years. Considering depression screening programs implemented at systematic levels, it remains unclear to what extent questions related to alcohol misuse were included in screening tools. Therefore, identification and referral to treatment for co-occurring MDE and AUD may not have kept pace with increased assessment and treatment of MDE. While CRAFFT—the most widely used screening instrument for alcohol and substance misuse for adolescents—is already being implemented in primary care settings since being endorsed by the American Academy of Pediatrics (Mitchell et al., 2014), more implementation research is needed to explore existing barriers to its use in tandem with depression screenings across other settings including university campuses. There is also moderate, but promising evidence, suggesting behavioral and psychosocial therapies like combined Motivational Interviewing (MI) and Cognitive Behavior Therapy (CBT) can improve both depression and drinking outcomes (Riper et al., 2014), and that integrated CBT can improve alcohol misuse and depressive symptoms just as well as, or better than, 12-step programs (McHugh & Weiss, 2019). Emerging evidence suggests certain combinations of MDE and AUD medications (e.g., sertraline with naltrexone and escitalopram with acamprosate) are associated with positive outcomes, including safety and reduction of both symptom types (Pettinati et al., 2010; Witte et al., 2012). Knowledge about effectiveness of integrated care can help inform development of interventions to engage young adults with co-occurring MDE and AUD into treatment.

The gap between the need for co-occurring MDE and AUD treatment further pinpoints the insufficiency in mental health workforce as a public health crisis. Through routine checkup, for example, primary care physicians are on the frontline of recognizing and initially addressing substance use disorders in young adults; yet, most internal medicine residents report they feel unprepared to treat addiction (Wakeman et al., 2013). Consistency of training in substance use disorders for medical residents has not been achieved, with the percentage of required training in programs at only 56% (Isaacson et al., 2000). While recent milestones including development of an addiction medicine specialty in 2015 by the American Board of Medical Specialties indicate some progress (Ayu et al., 2015; Wood et al., 2013), physicians who are certified in addiction medicine mostly work in specialty addiction medicine settings, rather than primary care settings. Major obstacles continue to exist

for training physicians, including lack of time, limited workforce to train, and low incentives for an additional year in fellowship training (Lembke & Humphreys, 2018).

Such a dearth in workforce is also seen among social workers, who provide more mental health services than all other related professions combined and work across diverse settings beyond specialty addiction facilities (Wells et al., 2013). Indeed, needs assessments identify multiple barriers to addiction training within social work programs including lack of available credits, developed course materials, and faculty equipped to teach content (Minnick, 2019). These systemic barriers confirm the need to harness online platforms in promising web-based substance use curriculums such as Kognito, which implements tenets of evidence-based Screening Brief Intervention and Referral to Treatment (SBIRT), and has already been studied among various healthcare providers (Bernstein et al., 2007; Koetting & Freed, 2017; Ryan et al., 2018; Tanner et al., 2012). Future studies investigating such virtual SBIRT training across programs of health care professions are further justified by its cost effectiveness, wide accessibility, and added convenience.

There is also urgent need to improve multidisciplinary service care systems for better service coordination. One issue is lack of consensus in terminology and conceptual frameworks across health disciplines and many providers still adopt the sequential model where a person is treated for one condition, then the other (Horsfall et al., 2009). Funding and structural barriers must be addressed to promote coordinated and integrated service provision, improved management of staff resistance to change, case management services in the community and expanded training to providers, including implicit bias when working with substance using patients (Ziedonis et al., 2005).

Disparities in Treatment Use for Co-Occurring MDE and AUD.

As we hypothesized, extensive differences were observed in both the prevalence and treatment use of MDE, AUD, and co-occurring conditions by young adults' sociodemographic characteristics. Specifically, despite higher prevalence of MDE, lower rates of treatment use were found in younger adults 18–21 years of age compared to those ages 22–25. It is possible, though, that instead of using professional services, younger adults might use digital-based programs and mental health promotion apps more often, which are designed to address a range of other health behaviors (Chandrashekar, 2018; Kroeze et al., 2006). Ninety-one percent of adults aged 16–24 reportedly use social networks on smartphones and are more likely to use the internet to seek information about mental health issues than from healthcare professionals (Alvarez-Jimenez et al., 2020). It is feasible, however, that because 18–21-year-olds are more likely to be dependents under parental healthcare plans than their 22–25-year-old counterparts, the anonymity offered in alternative digital-based interventions may provide greater incentive to seeking help for MDE. Likewise, lower rates of treatment use for AUD was noted in this younger cohort. One plausible explanation for lower AUD treatment among 18–21-year-olds is they may not perceive a need for AUD treatment in the same ways as 22–25-year-olds may, due to variations in peer influence and reinforcement within their social environment (Patrick & Maggs, 2011). In contrast, perceived need for treatment becomes harder to deny with increased age, due to worsening symptoms and cumulative adverse consequences over time, such

as mandated treatment from courts. Additionally, although campus services are a salient healthcare setting for 18–21-year-olds, previous research suggests that skepticism on treatment effectiveness and a general lack of perceived urgency deter college students with mental and behavioral health problems from seeking treatment on campus (Eisenberg et al., 2011). It is also important to put a caveat here that, in 2019, only 41% of 18–24-year-olds were enrolled in college (National Center for Education Statistics [NCES], 2021). To better address the unmet treatment needs of young adults with MDE and AUD, future studies with a focus on examining patterns across settings of service provision are essential.

Gender disparities revealed that females used more MDE treatment but less AUD and co-occurring treatments than males. Epidemiological evidence suggests females consistently received more professional treatment and prescription drugs for depression than did their male counterparts (Angst et al., 2002). Moreover, health providers may initiate treatment for females who have co-occurring disorders with greater efficiency than males as a result of more gender-based coordinated services that exist for females (Satre et al., 2010). Studies also indicate males may be more likely to report side effects of antidepressant medication use than females (Kornstein et al., 2000) as well as psychological side effects from diagnostic labeling or frequent symptom monitoring (Pyne et al., 2003). These cumulative findings point to the need for gender-specific outreach and treatment engagement interventions for males. Nonetheless, females between 18 and 25 have been underrepresented in research on co-occurring MDE and AUD, suggesting that it is essential to enhance understanding of gender differences and mechanisms underlying their heightened vulnerability for depressive disorders and alcohol misuse, such as disempowering gender norms, disproportionate child-rearing responsibilities, and premenstrual dysphoric disorder (McHugh & Weiss, 2019).

Surprisingly, young adults from families with higher household income were less likely to use co-occurring treatments than those with lower incomes. It is plausible that higher incomes afford individuals with greater social capital to navigate alternative coping strategies such as recreation and vacations, holistic interventions like functional medicine and spiritual retreats, and fitness and diet consultants; the financial option in hiring legal representation to hedge court mandated treatment; or the economic flexibility to leave stressful situations such as unfulfilling jobs or abusive relationships. The present study found that having health insurance increased the use of MDE treatment, but did not increase treatment use for AUD or co-occurring MDE and AUD, among those with both conditions. This may be a consequence of the prevailing belief that substance use disorders including AUD are primarily social rather than medical problems, which leads most primary care settings devoid of addiction screening, diagnosis, brief interventions, or medication management needed to address the prevalence seen in patients (McLellan et al., 2000). Another possible explanation is that integrated care are not included in all healthcare plans; instead, patients need to obtain referrals from primary care providers to access AUD treatments. This may serve as another obstacle, whereas primary care providers are more readily equipped to prescribe MDE medication without the need to refer out. Still, one study among adults over age 17 examining the role of insurance in utilization of alcohol and drug treatment found that insurance coverage played no significant role in the likelihood of treatment use, where entry to treatment was facilitated by ultimatums from the legal system or workplace

rather than from physicians or family (Weisner et al., 2002). These findings point to the “rock bottom” nature of the pathways to AUD treatment that may consistently reveal treatment gaps for those identified as having AUD, given the extremely difficult nature of recovery.

Consistently, young adults with MDE-related severe functional impairment in different life domains—including home management, ability to work, close relationships, and social life—were more likely to use treatment for any of the three conditions (MDE, AUD, and co-occurring disorders). One study found that individuals with co-occurring MDE and AUD had developed MDE and behavioral problems before age 13 more frequently than those who did not have co-occurring MDE and AUD, and they also reported more suicidal behavior (Sher et al., 2008). This highlights an opportunity for early prevention and screening among young adults to recognize warning signs of depression and associated risks of alcohol to cope, generating more awareness to normalize mental health and addiction dialog and practice among peers.

Implications for Research and Practice. Widening disparities in co-occurring treatments utilized among young adults ages 18–25 with MDE and AUD are perpetuated by gender, emerging adulthood and socioeconomic status, warranting immediate action. These study findings point to the need for expanded services and coordination between mental health and addiction systems that are more critical to reexamine now—more than ever—in the ongoing coronavirus disease (COVID-19) pandemic. Indeed, the pandemic served to exacerbate risk factors for alcohol misuse including social isolation, grief and loss, economic disruption (Sugarman & Greenfield, 2021) and preexisting risk factors for substance use in adolescents and young adults (Sarvey & Welsh, 2021) as well as compromised existing treatments disrupted by COVID-19 restrictions. Emerging evidence demonstrates telehealth for co-occurring treatment of young adults during the COVID-19 pandemic led to increased engagement with youth who had competing time demands or who lived far from facilities (Bagley et al., 2021). Further attention to research on efficacy of telehealth-based treatments particularly for co-occurring MDE and AUD among young adults is critical. Indeed, studies of web-based depression interventions for youth and young adults using CBT and Solution-Focused Brief Therapy (SFBT) demonstrate potential to mitigate depressive symptoms and barriers to treatment by providing anonymity and compatibility with digital lifestyles (Kramer et al., 2014; Richardson et al., 2010).

Destigmatizing AUD treatment and increasing awareness of its consequences and etiology in co-occurring conditions deserve urgent public health attention targeted to young adults. Expanding harm reduction programs for this population, promoting less consequential drinking behaviors, and increasing referrals to peer recovery programs are essential to mitigate AUD beyond 12-step abstinence models. Such models can be effective temporarily but lack social congruence with this life course stage as a consequence of differing intrinsic and extrinsic motivating factors. Promoting prevention of AUD with brief interventions like SBIRT in primary care and schools may help young adults receive needed attention for problematic behaviors before reaching more extreme consequences. Existing MDE treatments should also routinely incorporate assessment and interventions for AUD, and likewise, AUD treatments should provide therapies that address the common co-occurrence of MDE.

Limitations of the Study. The study was based on NSDUH datasets employing self-reports which may not correspond with clinical diagnosis. Nevertheless, while variations in sources of reporting would likely affect prevalence estimates of disorders, it is unlikely to affect temporal trends on which this study focused. Moreover, self-selection bias or social desirability cannot be ruled out, despite high response rates and attention to confidentiality. Third, the NSDUH does not recruit incarcerated, homeless, or military personnel and thus may restrict more comprehensive understanding of prevalence and treatment use trends among young adults. Additionally, this annual cross-sectional study did not use a longitudinal design and consequently these findings did not examine changes in individual level prevalence and treatment use over time.

Conclusion

To our knowledge, this study is the first to analyze temporal trends in prevalence of co-occurring MDE and AUD and treatment use patterns among young adults 18–25 years based on a large nationally representative sample. Understanding the unique needs of this emerging adult population requires innovative training and practice models to mitigate treatment gaps and widening disparities for co-occurring MDE and AUD. Because the etiology of these co-occurring conditions among young adults remains unclear, more longitudinal studies are needed to examine these explanatory associations, including the effect of gender, race, and socioeconomic status on these associations. The steady prevalence of these conditions among young people in light of staggering unmet treatment needs is a public health crisis that calls for wider implementation of accessible harm reduction programs on university campuses and in primary care settings, ramping up workforce strategies through systemic trainings, and digital or telehealth interventions for broader outreach.

Keywords: depression, alcohol disorder, co-occurring depression and alcohol misuse, young adults, co-occurring treatment

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